



School of Informatics & IT
TEMASEK POLYTECHNIC

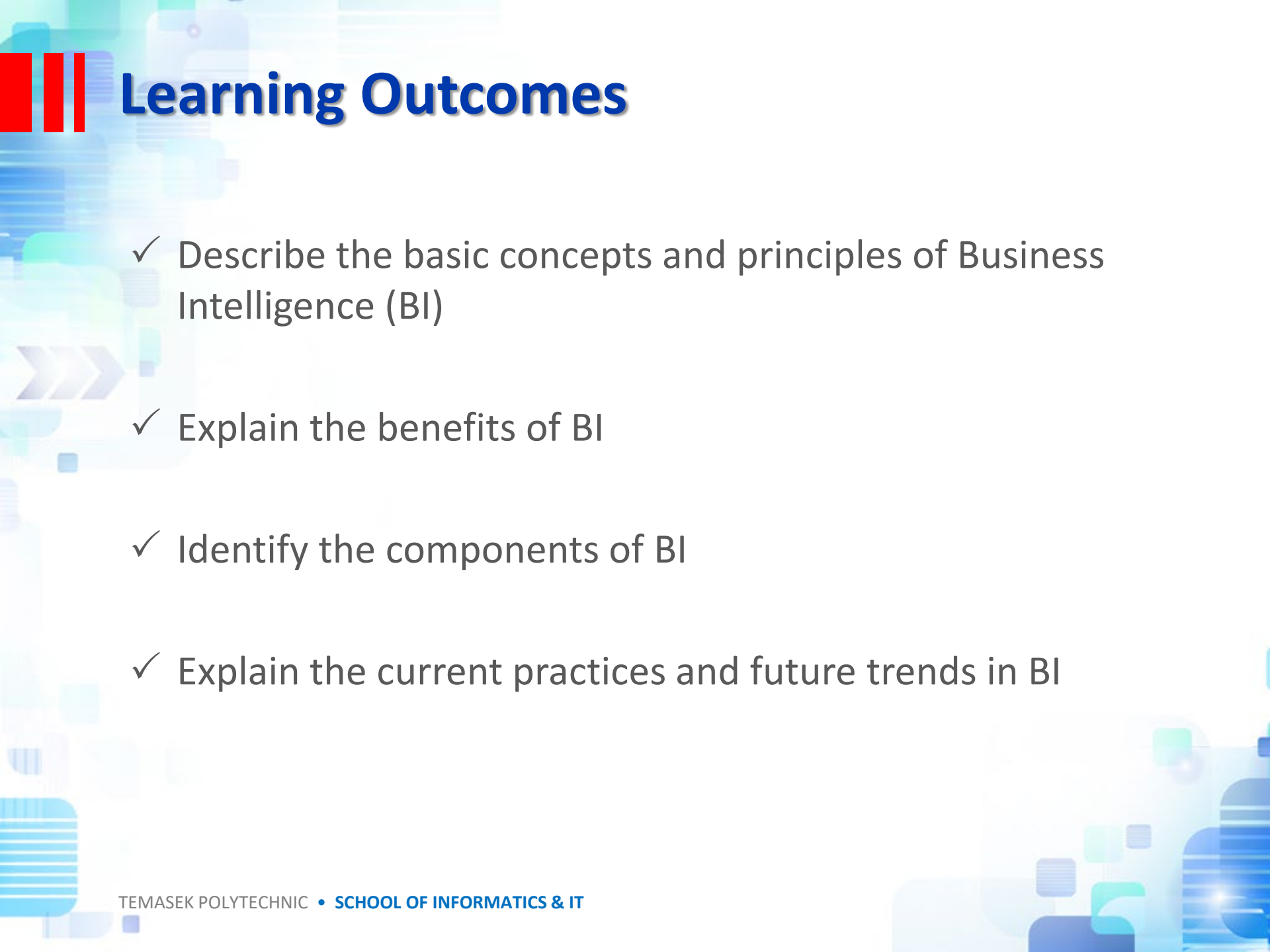
Data Warehousing and Business Intelligence

CDA2C01



Topic 1

Concepts and Principles of BI



Learning Outcomes

- ✓ Describe the basic concepts and principles of Business Intelligence (BI)
- ✓ Explain the benefits of BI
- ✓ Identify the components of BI
- ✓ Explain the current practices and future trends in BI

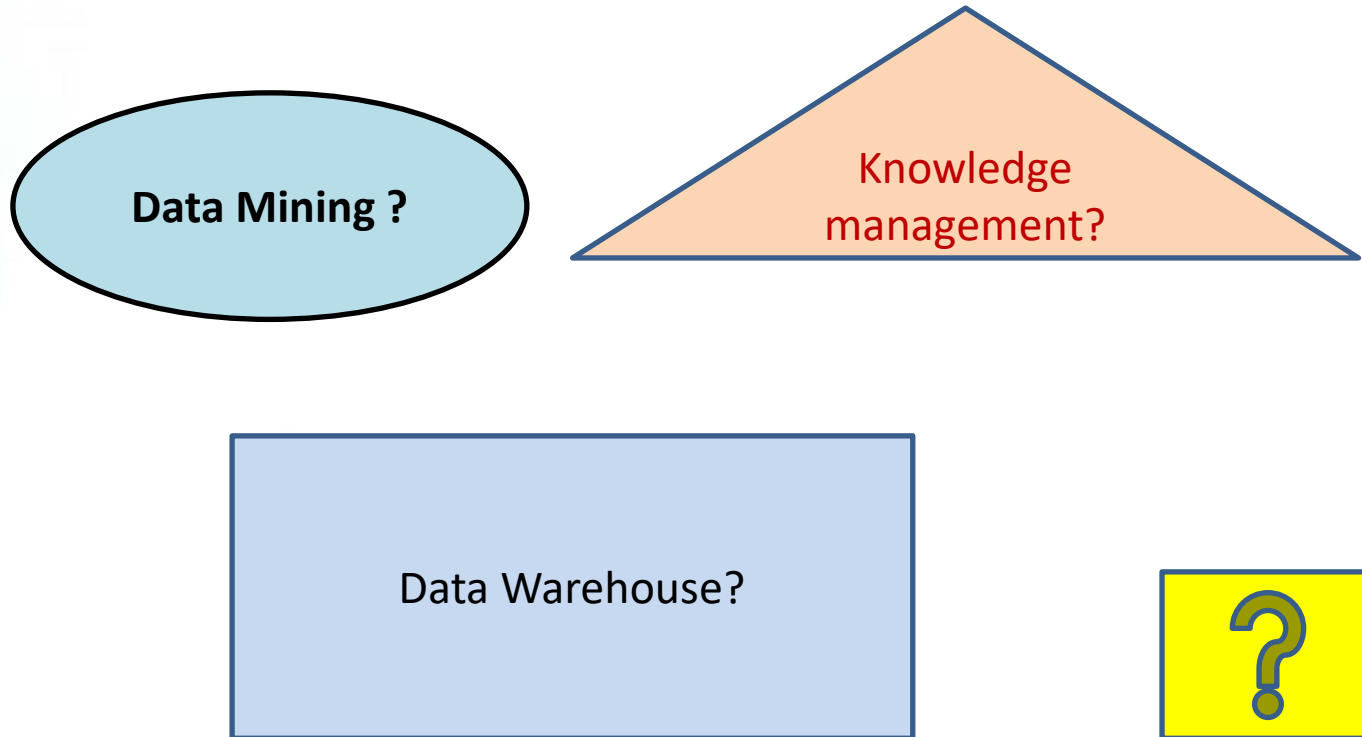
Brief History of Business Intelligence (BI)

- In 1958 , IBM researcher *Hans Peter Luhn* used the term **business intelligence**. He defined intelligence as: "the ability to apprehend the interrelationships of presented facts in such a way as to guide *action* towards a *desired goal*."
- During the 1980s, BI evolved from decision support systems, executive information systems and data warehouse.

Brief History of Business Intelligence (BI) (Cont'd)

- In 1989, Howard Dresner from Gartner proposed "business intelligence" as an umbrella term to describe "concepts and methods to improve business decision making by using fact-based support systems."
- Nowadays, we have Business analytics, business discovery and Big Data etc.

Various Definition of BI?



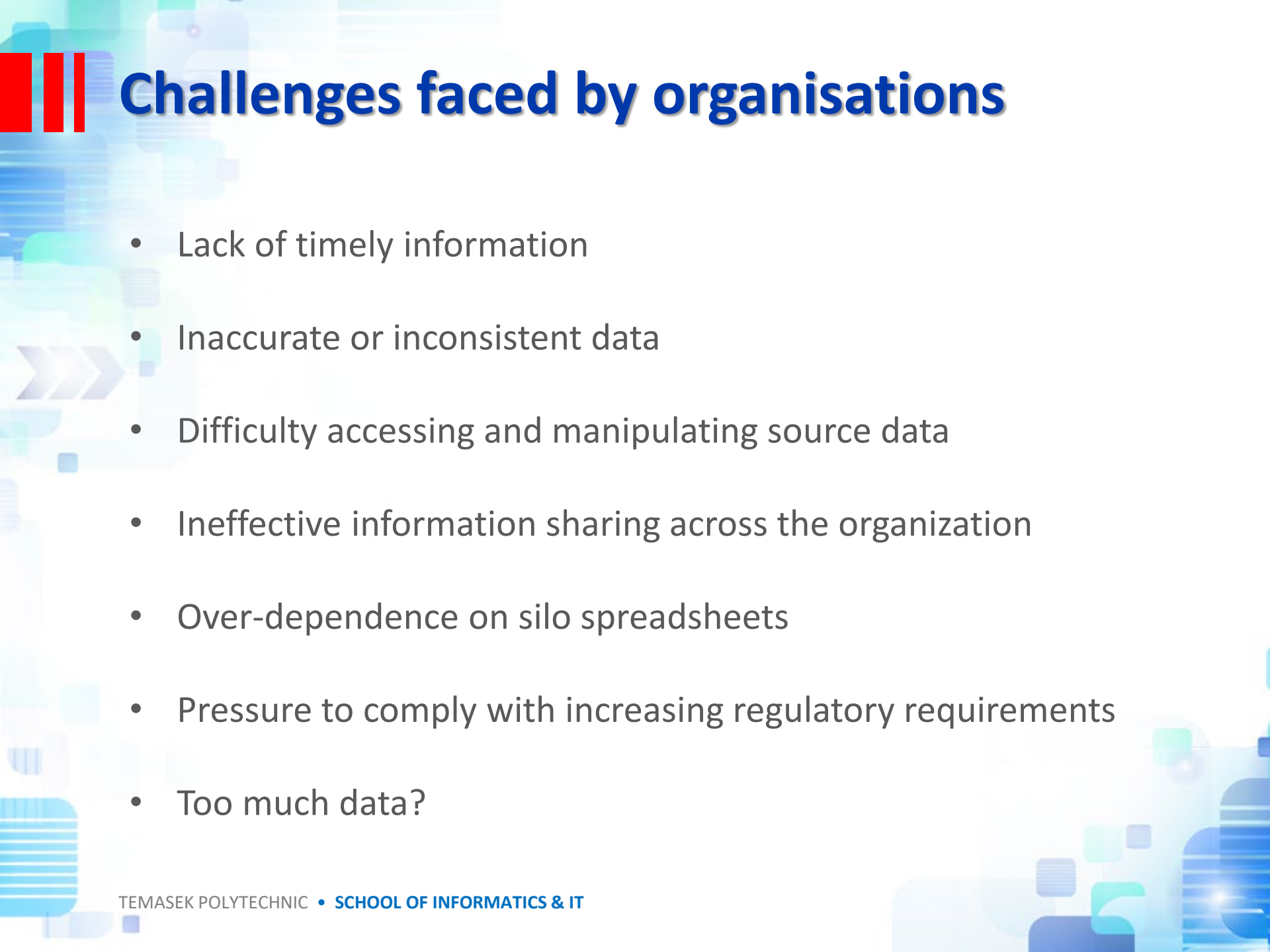
- Reports? Dashboard? Scorecard?

What is business intelligence?

- ***Business intelligence*** is an umbrella term that includes the applications, infrastructure and tools, and best practices that enable access to and analysis of information to improve and optimize decisions and performance (Gartner).
- Forrester defines the ***business intelligence (BI)*** market as a set of methodologies, processes, architectures, and technologies that leverage the output of information management processes for analysis, reporting, performance management, and information delivery. Research coverage includes executive dashboards as well as query and reporting tools.

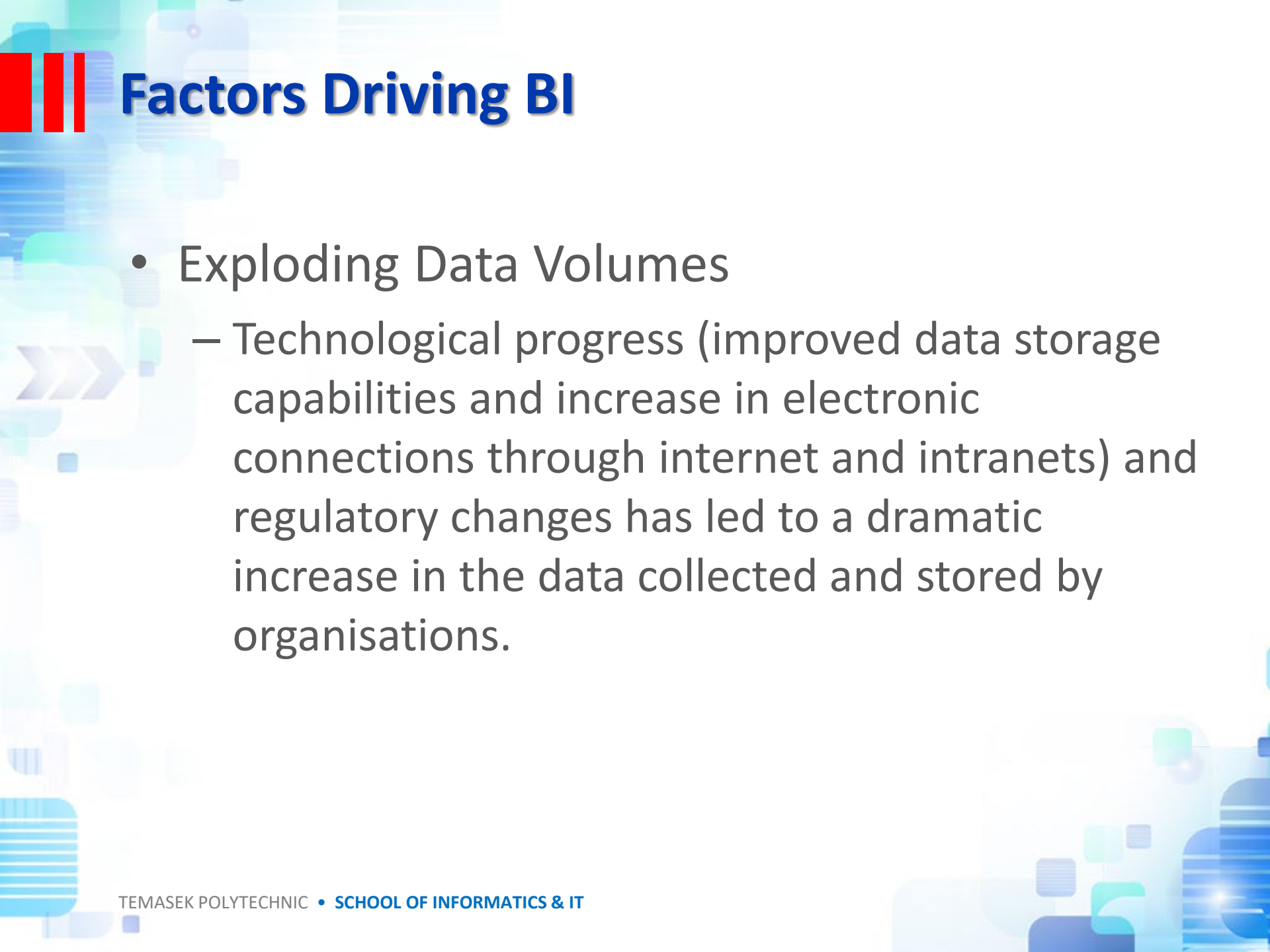
What is business intelligence?

- *Business Intelligence* is a user-oriented process of gathering and analyzing of data, which leads to the streamlining and rationalization of the decision-making process (Jerzy Surma).



Challenges faced by organisations

- Lack of timely information
- Inaccurate or inconsistent data
- Difficulty accessing and manipulating source data
- Ineffective information sharing across the organization
- Over-dependence on silo spreadsheets
- Pressure to comply with increasing regulatory requirements
- Too much data?



Factors Driving BI

- Exploding Data Volumes
 - Technological progress (improved data storage capabilities and increase in electronic connections through internet and intranets) and regulatory changes has led to a dramatic increase in the data collected and stored by organisations.

Factors Driving BI (Cont'd)

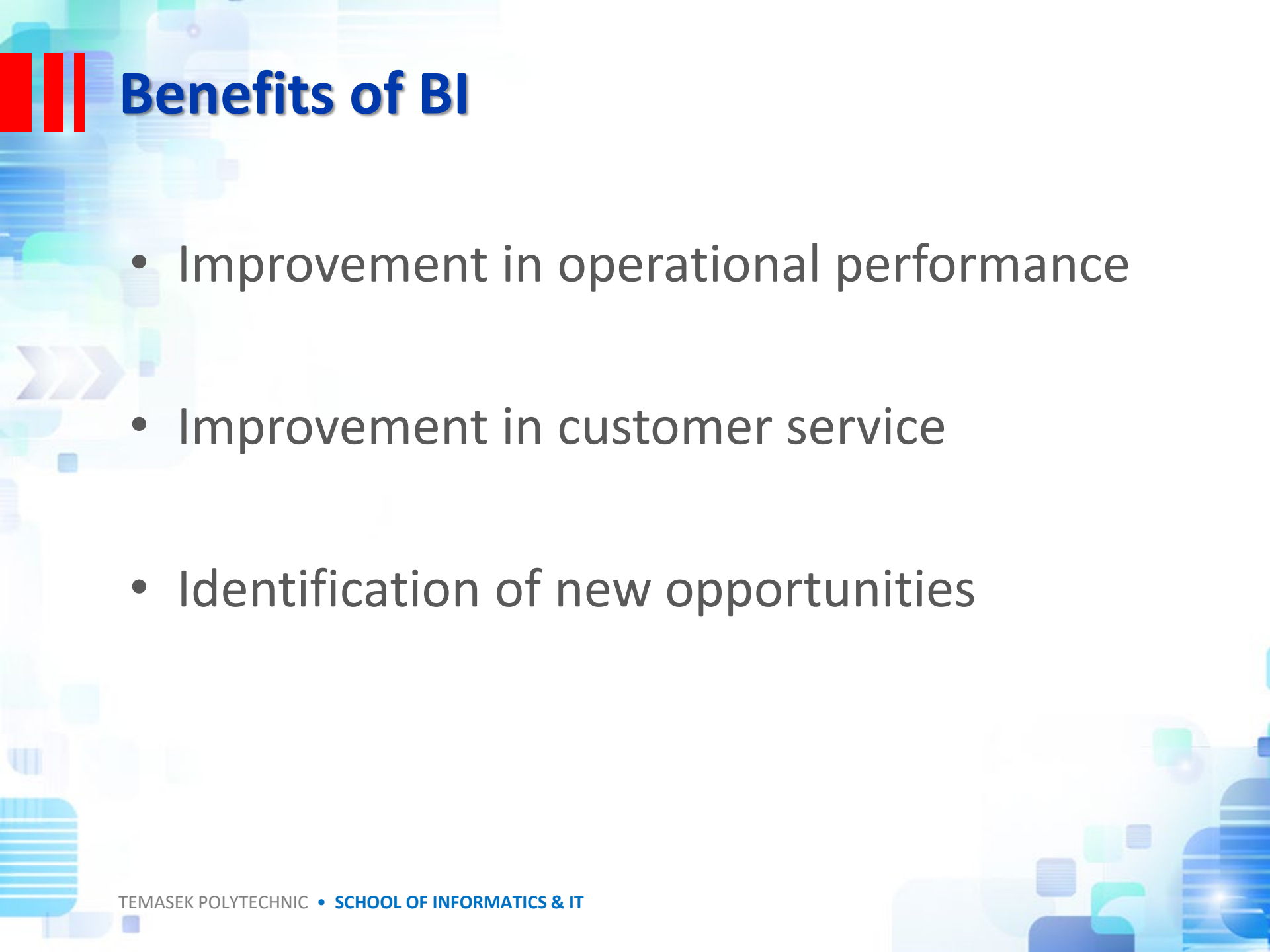
- Increasingly Complicated Decisions
 - With increased competition (across industries and countries), decision making has become increasingly complicated. The variety of factors that need to be considered and the availability of greater information make decision making more complex.

Factors Driving BI (Cont'd)

- Need for Quick Reflexes
 - The pace of change within each market domain has increased rapidly. The time available for organizations to respond to changes has been decreasing. Managers need to quickly access actionable information, so that decisions can be made and implemented swiftly.

Factors Driving BI (Cont'd)

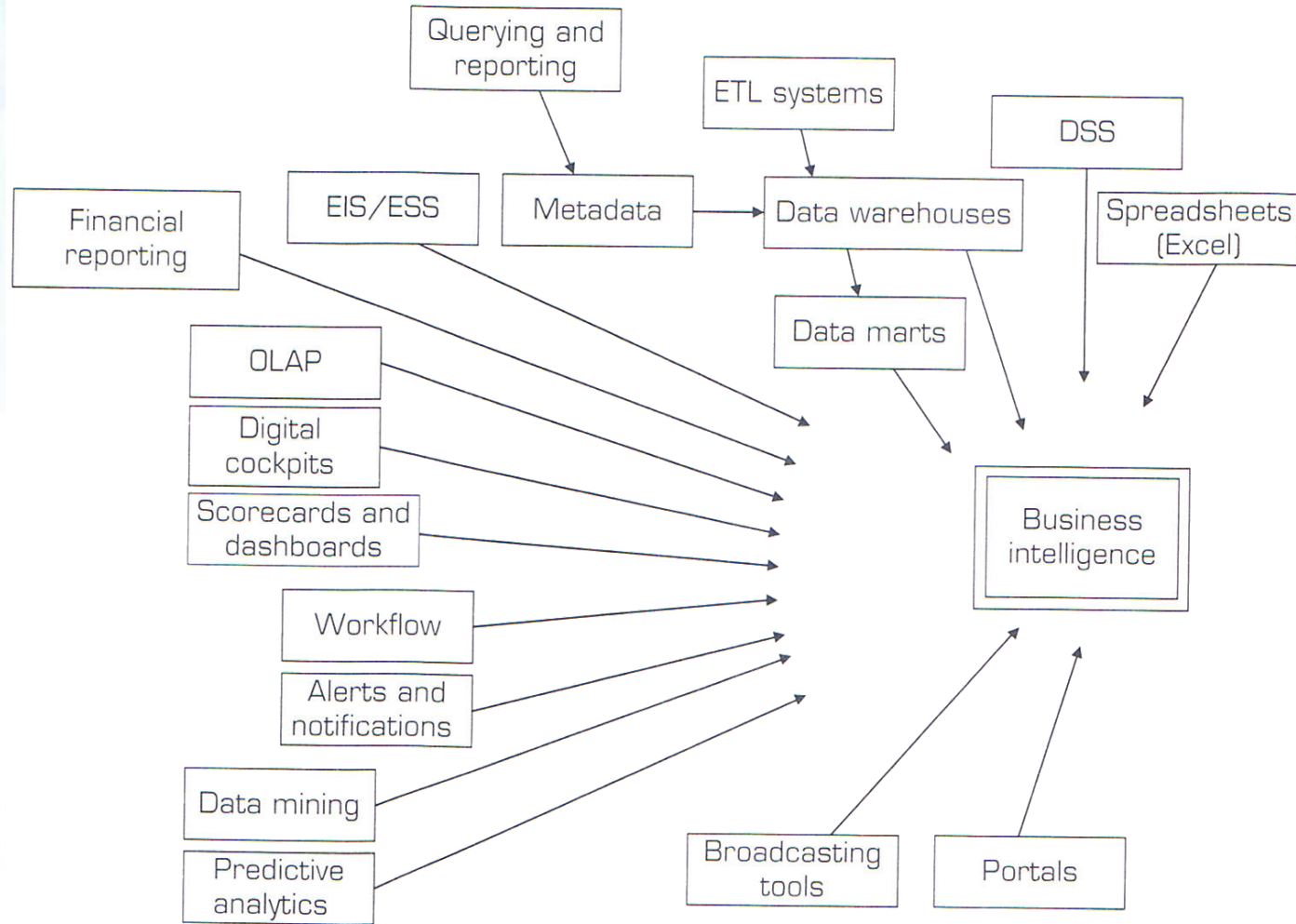
- Technological Progress
 - The utilization of BI in contemporary organisations is made possible by the developments in decision support systems, data warehousing, data mining and text mining. BI tools provide organisations with the platform to implement effective BI solutions.



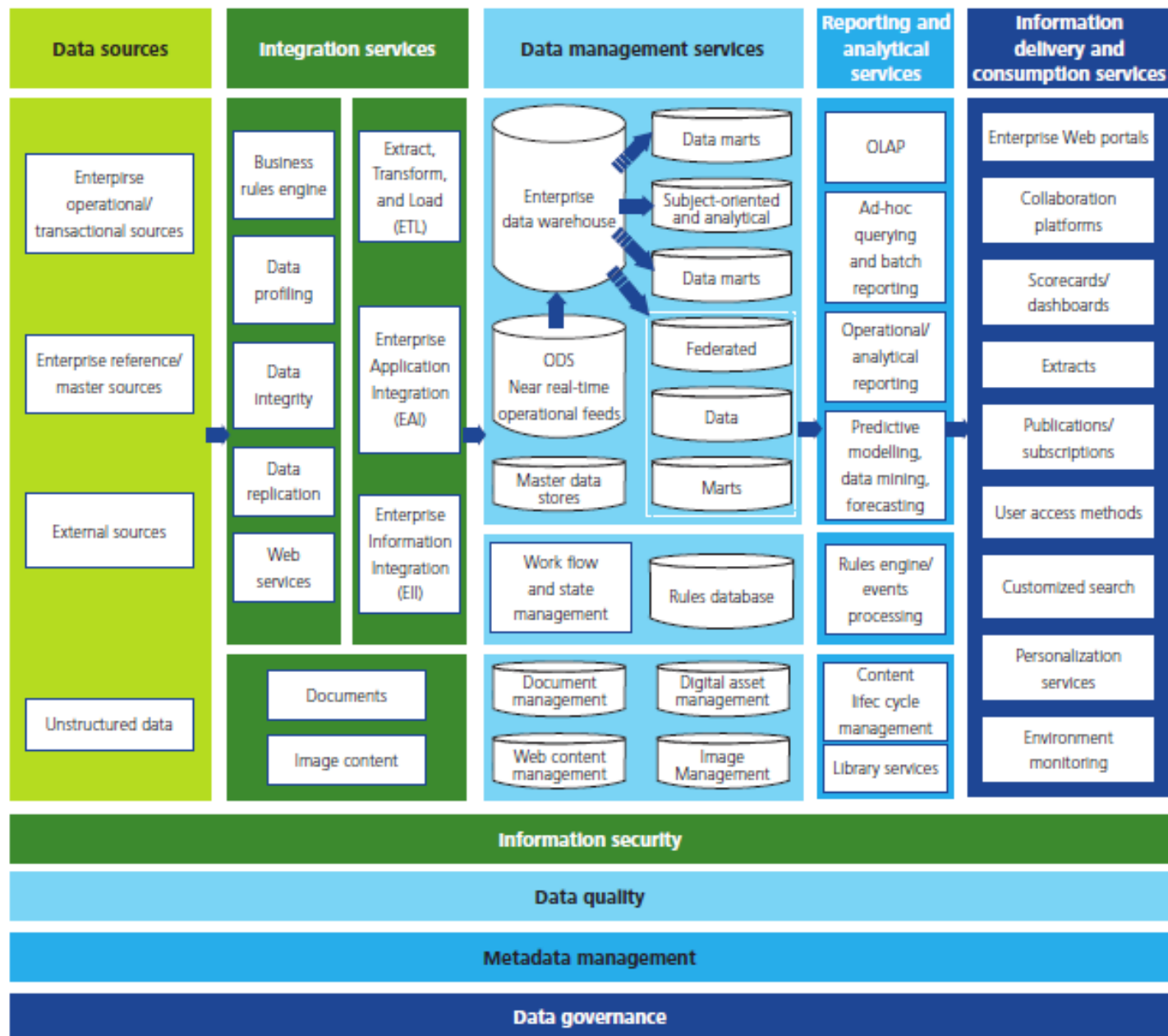
Benefits of BI

- Improvement in operational performance
- Improvement in customer service
- Identification of new opportunities

BI Evolution



Typical BI Architecture



Envisioned Solution Architecture



Business Analytics Framework

Enterprise Data Sources



Existing Systems



New Systems



Other RDBMS



Ad-Hoc Data

Data Integration

- ETL
- Data Access
- Impact Analysis
- Data Quality

Data Warehouse

Analytics

- Guided Analysis
- Statistical Analysis
- Data & Text Mining
- Forecasting

Enterprise Business

- Dashboards/Alerts
- Web Reports
- OLAP drill/slice
- MS Office Integration
- Visual BI
- Mobile BI

Central Metadata Management

BI user groups

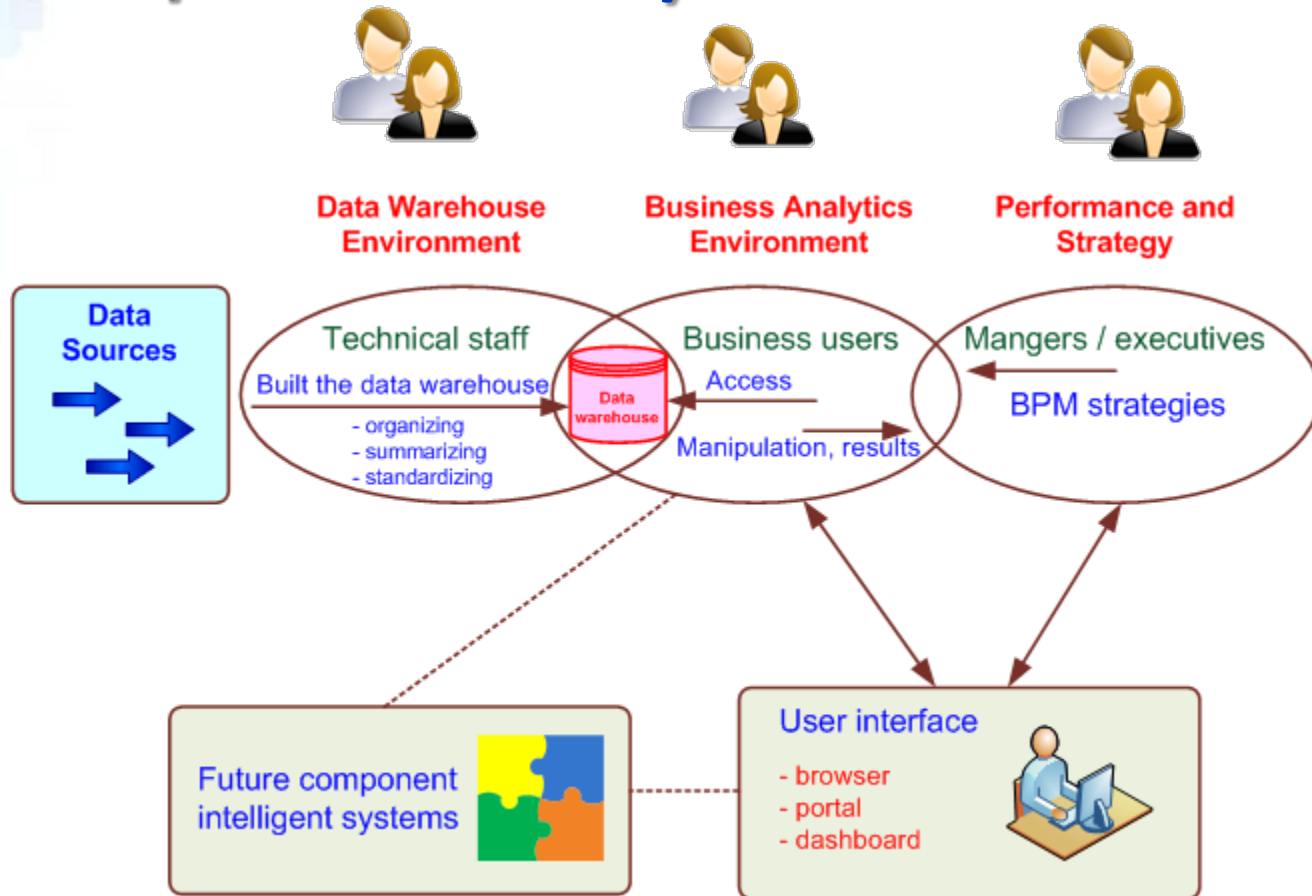
Internal Users

- Senior Manager
- Manager
- Analyst
- Frontline technical, clerical and sales staff

External Users

- Customer of products/ services
- Supplier of products/ services
- Information Consumer
- Regulators

A high-level architecture of BI (USERS / PERSONAS)



The architecture of BI depends on its applications.

Impact of BI

The Impact of Business Intelligence in an Organization

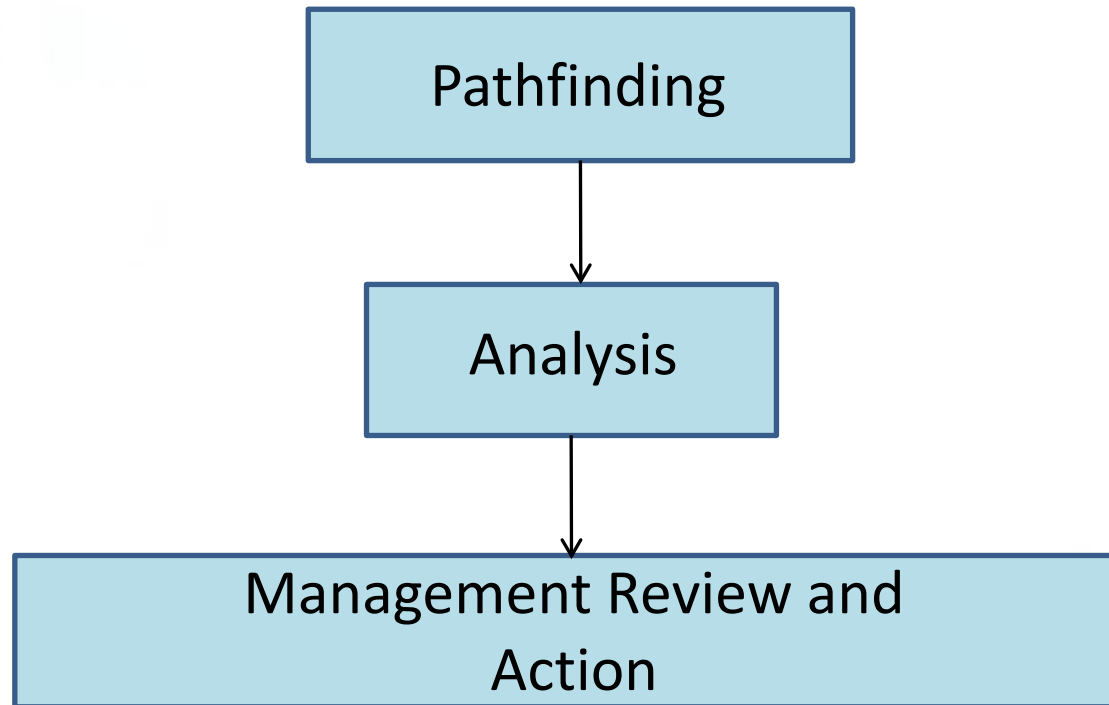
In planning for a BI solution, the project manager needs to understand the relationship between BI and the following user groups:

- The Information Technology (IT) department
- Non-Technical end-users
- Business Analysts
- External Users
- The Enterprise

The user's **PERCEPTION of the benefits received from BI** will be proportionate to the degree of how much productivity has increased or the amount of insights that they received.

Decision Making Process

- Decision making requires that three basic tasks be performed:



Decision Making Process - *Pathfinding*

- Pathfinding is the step at which possible acts that can be taken are identified. Pathfinding involves answering questions such as:
 - What alternatives (options) do we have?
 - If we have a problem, what can we do about it?
 - The organization is not doing as well as we feel it should; what can we change?
 - The competition is killing us. What can we do?
 - How do we get to where we want to be from where we are?

Decision Making Process - *Analysis*

- Analysis involves “laying out” the possible acts in some way, and comparing and contrasting them. Typical questions include:
 - What are the consequences of each act?
 - How can we choose one act over some other act?
 - What relevant data are available?
 - Is there some further data that we can collect?
 - Can expert opinion help us decide?
 - What “number crunching” or other analysis should be done?
 - Can some type of “model” help us decide?

Decision Making Process – *Managerial Review and Action*

- The manager must review both the pathfinding step and analysis step:

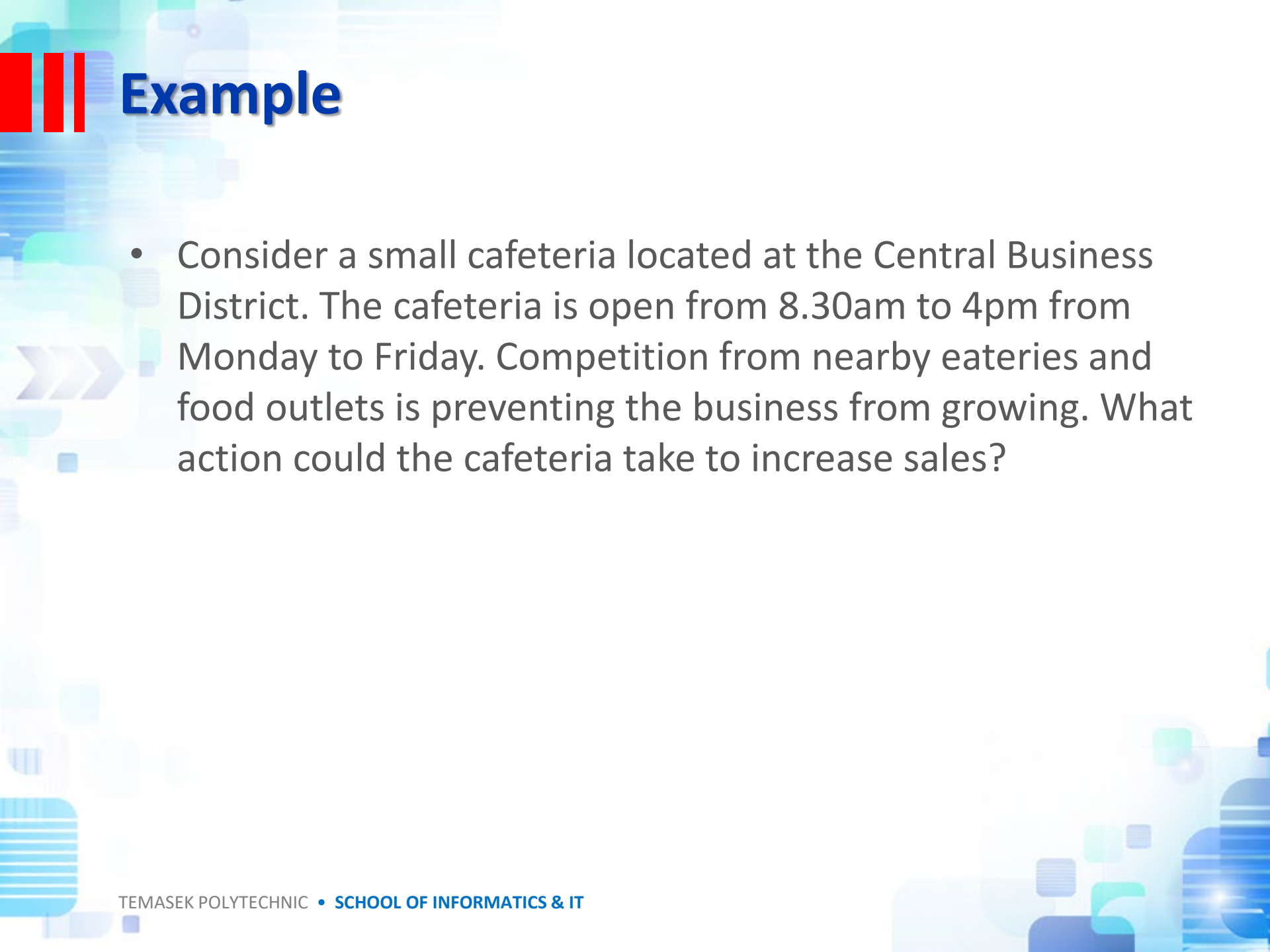
Pathfinding Step:

- Were the “right” actions considered?
- Were there others that were missed?

Analysis Step:

- Was the analysis valid?
- What conclusions were reached?
- What was learned from the analysis?

Fact-Based
Decision Making



Example

- Consider a small cafeteria located at the Central Business District. The cafeteria is open from 8.30am to 4pm from Monday to Friday. Competition from nearby eateries and food outlets is preventing the business from growing. What action could the cafeteria take to increase sales?

The Strategic/ Tactical/ Operational Framework

- Three types of decision making at different levels in the organization:



Decision Making Levels

- **Operational:**

- Takes place at the lower levels of the firm
- Required for day-to-day running of the organization
- Mainly short term consequences
- Can be changed fairly easily, often at low cost

Examples:

How much of a product to order when the stock has (or is about to) run out?

How much fuel to load into an airplane?

What route to take to deliver our packages?

Decision Making Levels (Cont'd)

- **Tactical:**

- Takes place in the middle, managerial levels of the organization.
- Have long-lasting effects than do operational decisions and are more costly to change.

Examples:

Deriving the manufacturer's master production schedule.

Deciding on the organization's operating budget.

Hire management staff.

Decision Making Levels (Cont'd)

- **Strategic:**

- Task of top management
- Affects long term direction and vitality of the organization.
- Generally, strategic decisions, once taken, are very costly and difficult to change.

Examples:

Leasing or buying a new building

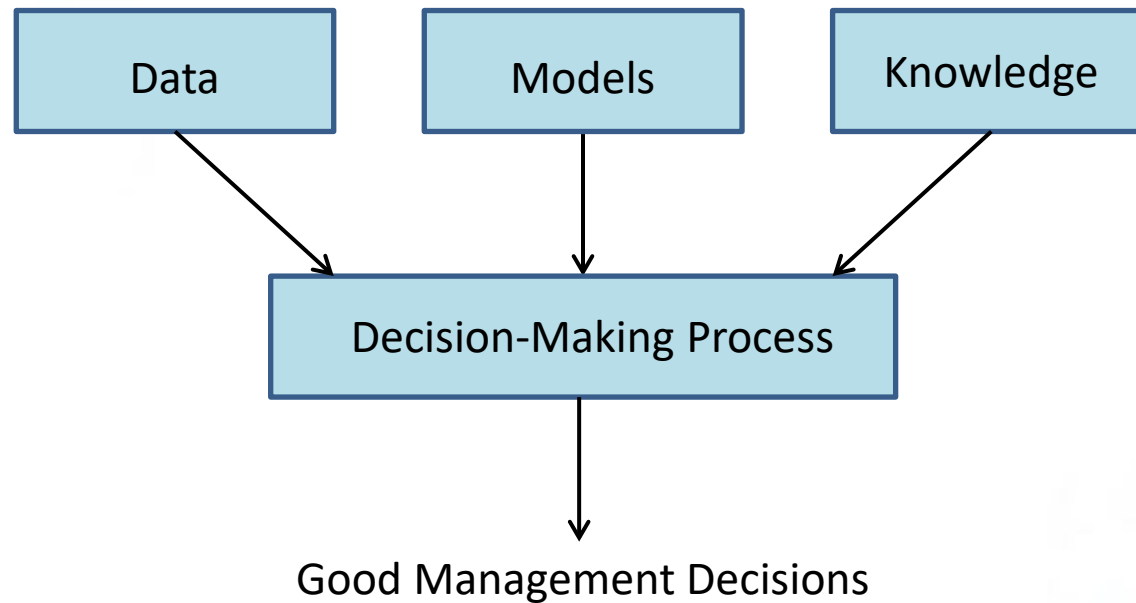
Launching a major new product

Acquiring another company

Selecting the CEO for the organization

Data, Models, and Knowledge (DMK) Framework - Inputs to a decision process

- The DMK framework characterizes the inputs in a decision process.



DMK Framework

- **Data**

- Key inputs to decision making
- Data can be specific to the organization (e.g. payroll numbers) or public (e.g. today's stock prices)
- Data can be captured in the organization's databases (in data warehouses), exists in the minds of employees, or is available externally (e.g. through public, electronic databases)

DMK Framework (Cont'd)

- **Models**

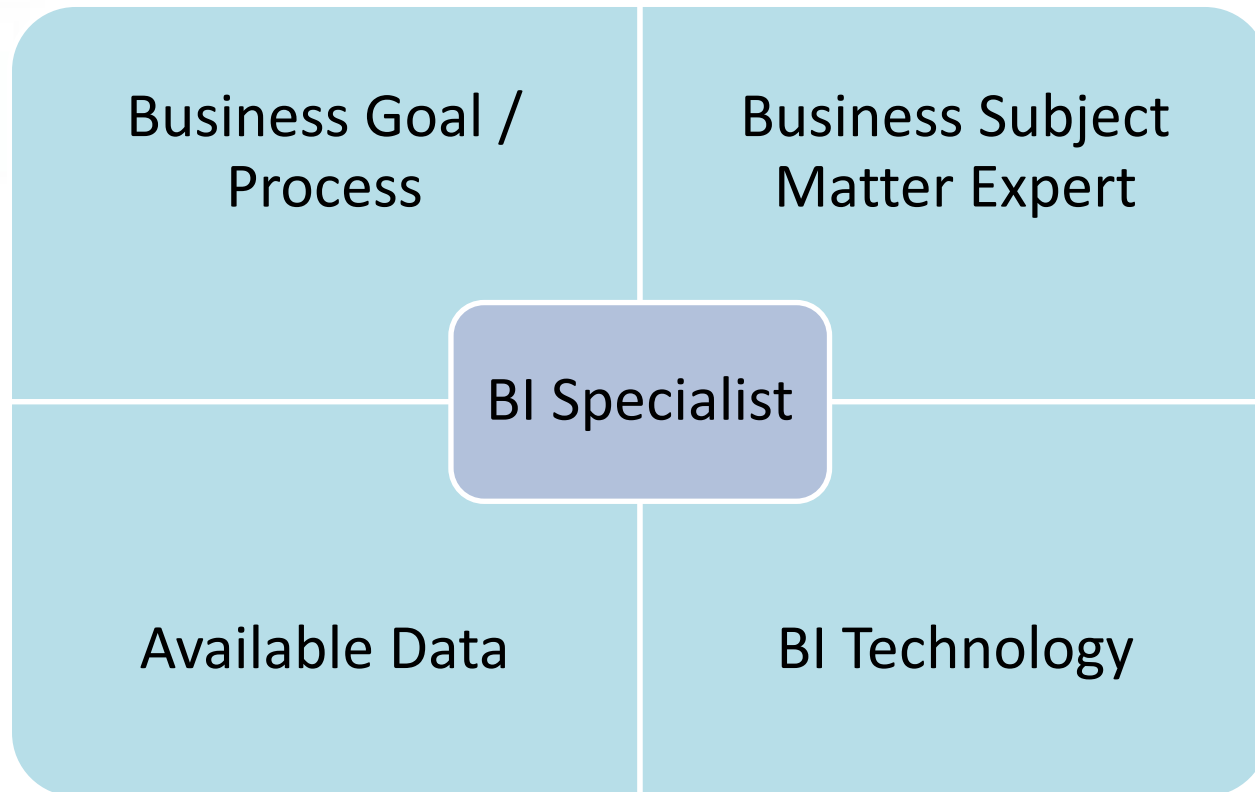
- Data must be compiled, abstracted, and manipulated in some way to be relevant to the decision. This is the role of model.
- A model is a construct that adds value to data.
- The use of model allows for an informed analysis of a decision situation, bringing relevant data and structure to the decision-making task.

DMK Framework (Cont'd)

- **Knowledge**

- Few decisions are made without knowledge.
- Vast amounts of knowledge resides in the minds of employees (mental models).
- Knowledge relevant to decisions comes from people involved in the decision-making process i.e. knowledge about data, knowledge about models and model building, and managerial knowledge about this decision and similar situations that have occurred in the past.

BI components



INFORMS – Three levels of analytics

- The Institute for Operations Research and the Management Sciences (INFORMS) defined three levels of analytics:
 - **Descriptive analytics:**
 - Prepares and analyzes historical data
 - Identifies patterns from samples for reporting of trends
 - **Predictive analytics:**
 - Predicts future probabilities and trends
 - Finds relationships in data that may not be readily apparent with descriptive analysis
 - **Prescriptive analytics:**
 - Evaluates and determines new ways to operate
 - Targets business objectives
 - Balances all constraints

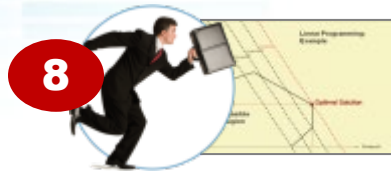
SAS - Eight Levels of Analytics

- Standard Reports
- Ad-hoc Reports/Scorecards
- Query/Drill Down
- Alerts
- Statistical Analysis
- Forecasting
- Predictive Modeling
- Optimization

Descriptive
Analytics

Predictive
Analytics

Prescriptive
Analytics



8

OPTIMIZATION

Answer the questions: How do we do things better?
What is the best decision for a complex problem?



7

PREDICTIVE MODELING

Answer the questions: What will happen next? How will it affect my business?



6

FORECASTING

Answer the questions: What if these trends continue?
How much is needed? When will it be needed?



5

STATISTICAL ANALYSIS

Answer the questions: Why is it happening? What opportunities am I missing?



4

ALERTS

Answer the questions: When should I react? What actions are needed now?



3

QUERY DRILLDOWN (OR OLAP)

Answer the questions: Where exactly is the problem? How do I find the answers?



2

AD HOC REPORTS

Answer the questions: How many? How often? Where?

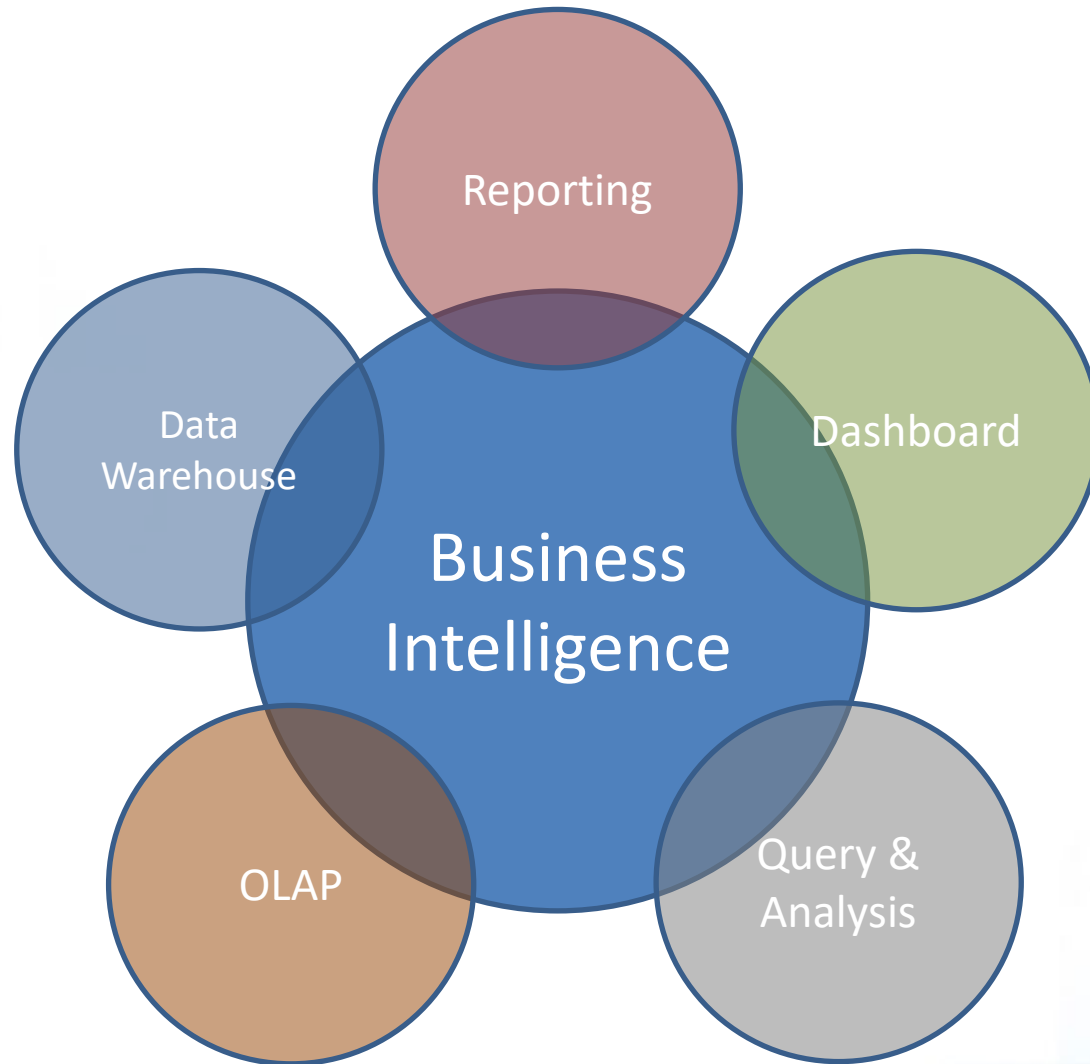


1

STANDARD REPORTS

Answer the questions: What happened? When did it happen?

Primary Components of BI



Reporting

- Reporting provides summarised views of the business.

Month to Date Report

Business Date: (28)
 CM Name: (28)

QuikServe SOLUTIONS

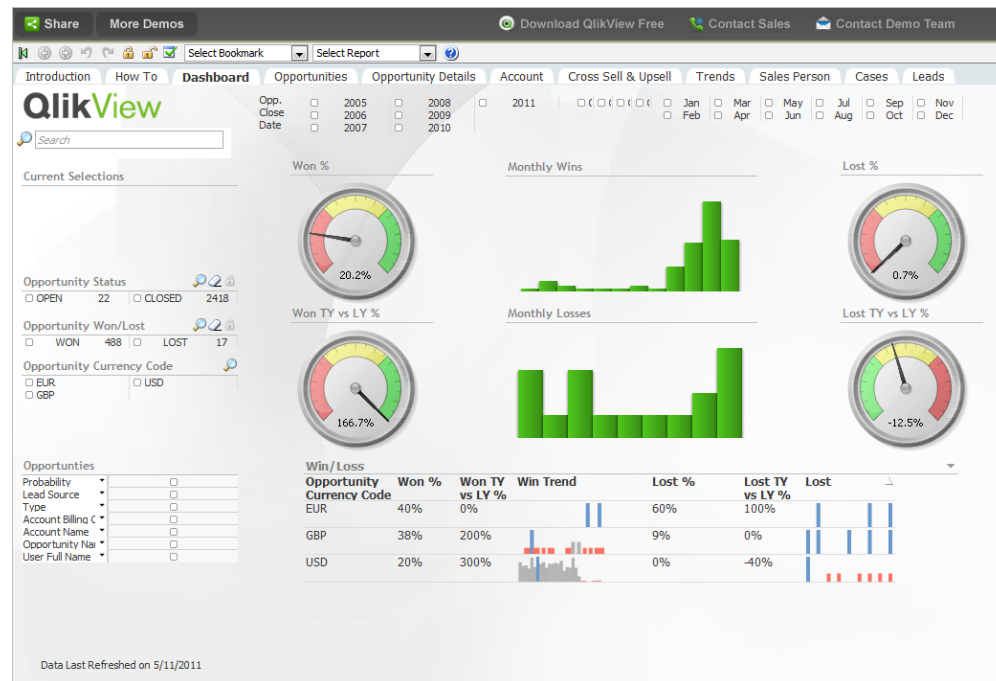
Rest	Adjusted Sales	Adjusted Sales LY	Sales %	Customer Count	Customer Count LY	Customer Count %	SOS Daypart 1	SOS Daypart 2	SOS Daypart 4	Total SOS	Cancel Item %	Cash OS	Employee Meals	Labor %	Labor Hours OS	Waste \$	Waste %	PLV Loss %
XXXX	31,819.19	27,617.83	15.21%	5620	5,002	12.60%	972	945	931	1031	7.47%	39.74	140.55	14.52%	(16.95)	149.34	6.47%	(1.50%)
XXXX	22,590.36	21,704.09	4.00%	2091	2,873	0.83%	795	1164	1204	779	6.25%	93.15	73.27	14.02%	(29.56)	94.04	6.42%	5.36%
XXXX	28,948.88	25,282.96	14.49%	3969	3,660	8.44%	822	881	970	890	10.04%	9.35	584.58	18.40%	54.30	148.70	6.60%	(0.37%)
XXXX	20,022.94	18,445.09	8.43%	2388	2,765	4.30%	745	828	1080	1050								
XXXX	23,386.75	21,442.81	9.12%	3443	3,318	3.77%												
XXXX	26,673.01	19,374.85	37.67%	3647	2,732	33.49%	929	723	1085	8								
XXXX	24,198.09	23,725.14	1.87%	3999	4,059	(2.44%)												
XXXX	37,484.59	34,329.38	9.20%	6095	5,038	1.13%	906	972	1082	10								
XXXX	36,325.43	32,713.17	10.97%	4779	4,195	13.92%												
XXXX	18,985.59	18,528.33	2.36%	2398	2,423	(1.03%)	982	1200	1305	11								
XXXX	35,587.20	31,552.27	12.79%	4477	4,285	4.48%												
XXXX	20,482.17	19,132.84	7.06%	2976	2,828	8.13%	1229	1223	1198	12								
XXXX	20,324.88	19,613.07	3.53%	3061	2,933	4.36%	832		904	5								
XXXX	38,111.55	43,654.44	(10.41%)	3979	4,914	(19.03%)												
XXXX	19,629.09	18,946.89	3.50%	3169	3,125	1.41%	737	793	907	8								
XXXX	18,449.34	16,624.13	10.98%	2982	2,833	5.26%	970	968	1164	7								
XXXX	26,111.73	23,797.06	9.73%	4235	3,844	10.17%	985	833	1797	11								
XXXX	30,757.58	28,182.94	9.14%	5059	4,985	4.61%	804	737	1780	10								
XXXX	18,037.71	15,950.31	13.02%	3439	3,208	7.20%	923	968	1241	9								
XXXX	19,873.59	16,969.25	16.91%	3591	3,192	12.57%	907	951	961	7								
XXXX	20,948.85	19,664.60	6.52%	3228	3,354	(3.76%)												
XXXX	29,338.07	26,915.74	9.00%	4014	3,720	7.90%	882	1551	1184	10								
XXXX	13,194.82	9,432.53	39.89%	1925	1,737	10.82%	692	775	974	8								
XXXX	19,307.83	16,085.07	20.04%	3119	2,730	14.25%	1038	1123	1510	12								
XXXX	23,291.54	19,648.24	18.54%	3419	3,146	8.60%												
XXXX	27,955.57	26,451.12	5.69%	3835	3,825	0.29%												
XXXX	25,216.22	26,068.95	(3.10%)	3740	3,937	(5.04%)	835	955	1376	10								
XXXX	23,250.01	26,765.69	(13.14%)	3182	3,685	(13.69%)												
XXXX	14,696.89	14,711.06	(0.08%)	2332	2,402	(2.83%)	1098	1286	1738	14								
XXXX	13,359.87	11,533.26	15.84%	2125	1,845	15.18%	1051	709	1189	8								
XXXX	26,452.41	27,089.97	(8.04%)	3731	3,961	(5.81%)												
XXXX	10,049.46	8,583.54	17.08%	1537	1,343	14.45%												
XXXX	25,529.48	22,850.79	11.84%	3445	3,068	12.29%	975	939	1147	8								
Grand Total	790,290.67	733,409.58	7.76%	115350	110,857	4.00%	20249	21410	26267	21								

Compare with Last Month by Categories

Category Name	Current Month		Last Month	
	Quantity	Amount	Quantity	Amount
Beverages	925	\$27,761.58	834	\$34,599.15
Condiments	378	\$10,773.27	289	\$6,293.97
Confections	880	\$22,877.18	475	\$10,074.09
Dairy Products	581	\$13,685.33	506	\$11,454.50
Grains/Cereals	189	\$3,325.40	219	\$4,060.01
Meat/Poultry	92	\$4,083.66	344	\$23,334.05
Produce	351	\$13,031.20	37	\$1,172.80
Seafood	669	\$9,316.55	584	\$12,531.12
Total	4,065	\$104,854.15	3,288	\$103,519.69

Dashboard

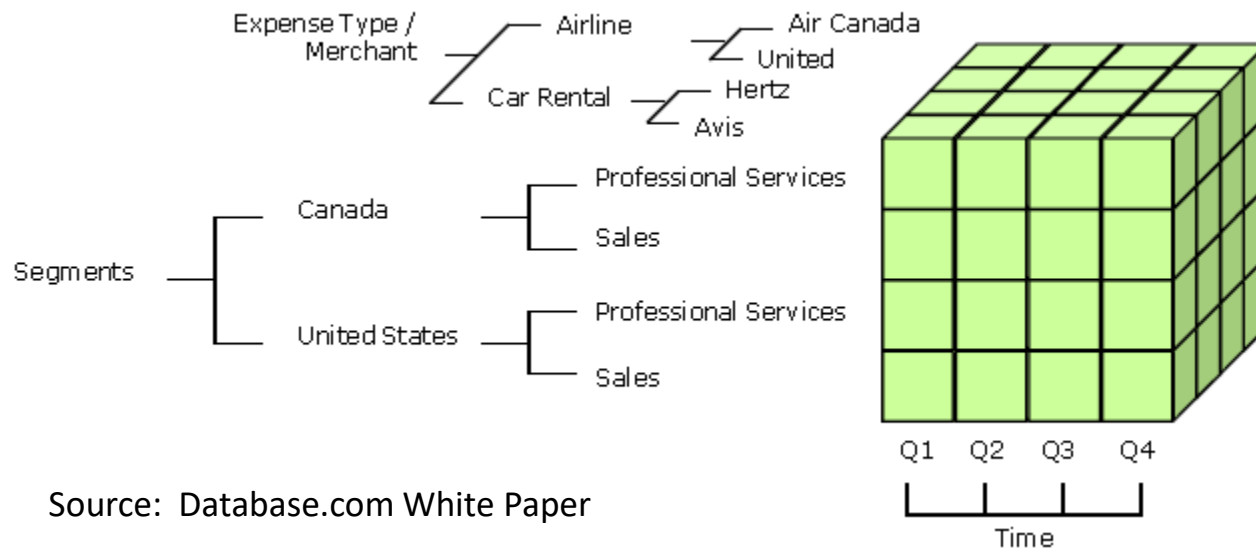
- A dashboard is a visual display of the most important information needed to achieve one or more objectives; consolidated and arranged on a single screen so the information can be monitored at a glance.



Data Warehouse

- Data warehouse (DW or DWH) is a system used for reporting and data analysis, and is considered a core component of BI. DWs are central repositories of integrated data from one or more disparate sources.

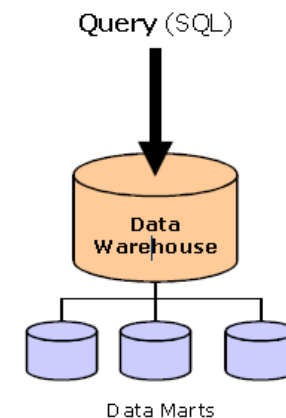
OnLine Analytical Processing(OLAP)



Source: Database.com White Paper

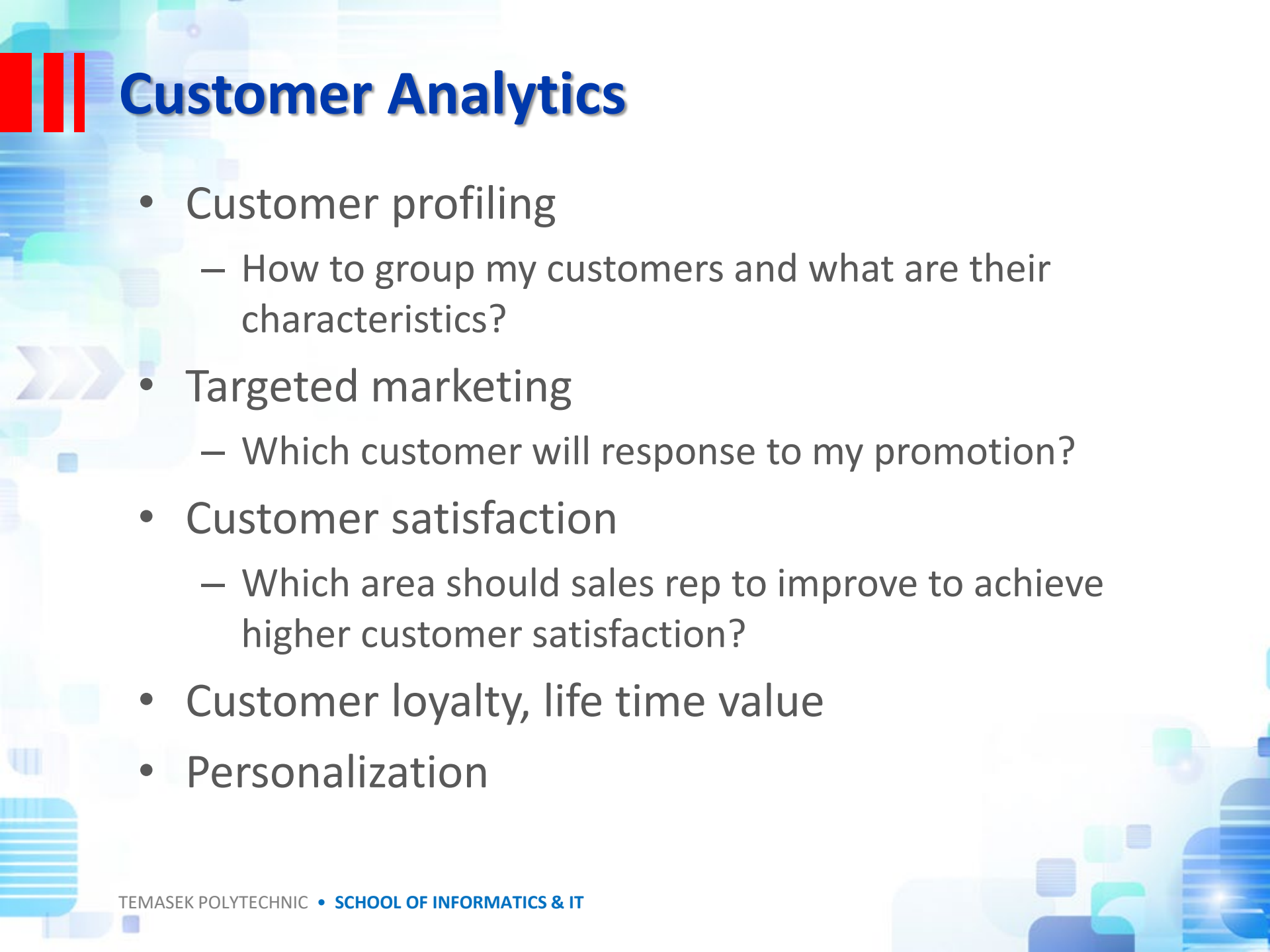
Query

- **Query**
 - The process of searching for a single file or collection of files to obtain lists of records that match the search criteria.
- **SQL (Structured Query Language)**
 - A computer programming language used to process data in a relational database.



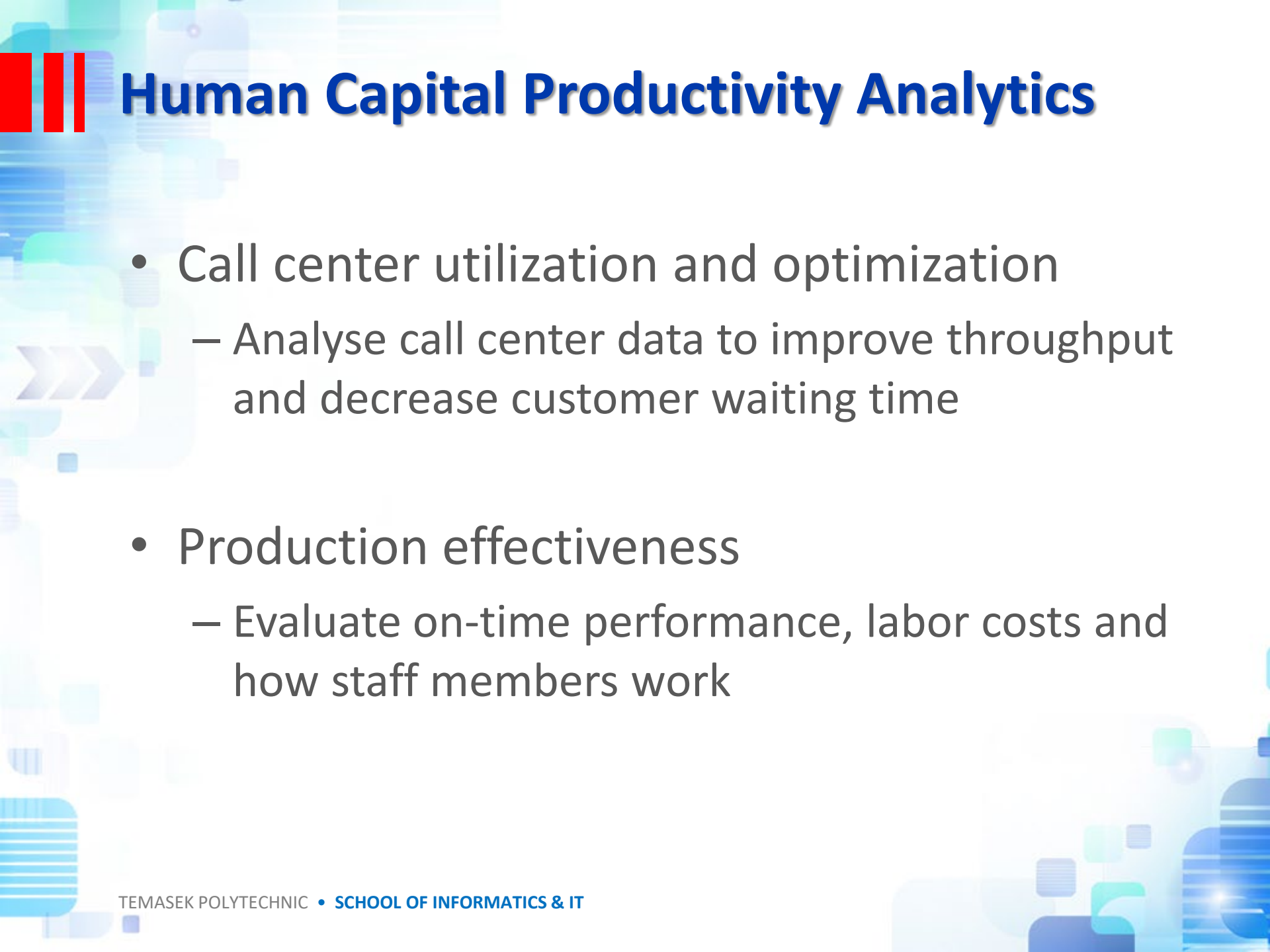
The Overview of BI application

Application
Customer Analytics
Human Capital Productivity Analytics
Business Productivity Analytics
Sales Channel Analytics
Supply Chain Analytics
Behavior Analytics



Customer Analytics

- Customer profiling
 - How to group my customers and what are their characteristics?
- Targeted marketing
 - Which customer will response to my promotion?
- Customer satisfaction
 - Which area should sales rep to improve to achieve higher customer satisfaction?
- Customer loyalty, life time value
- Personalization



Human Capital Productivity Analytics

- Call center utilization and optimization
 - Analyse call center data to improve throughput and decrease customer waiting time
- Production effectiveness
 - Evaluate on-time performance, labor costs and how staff members work

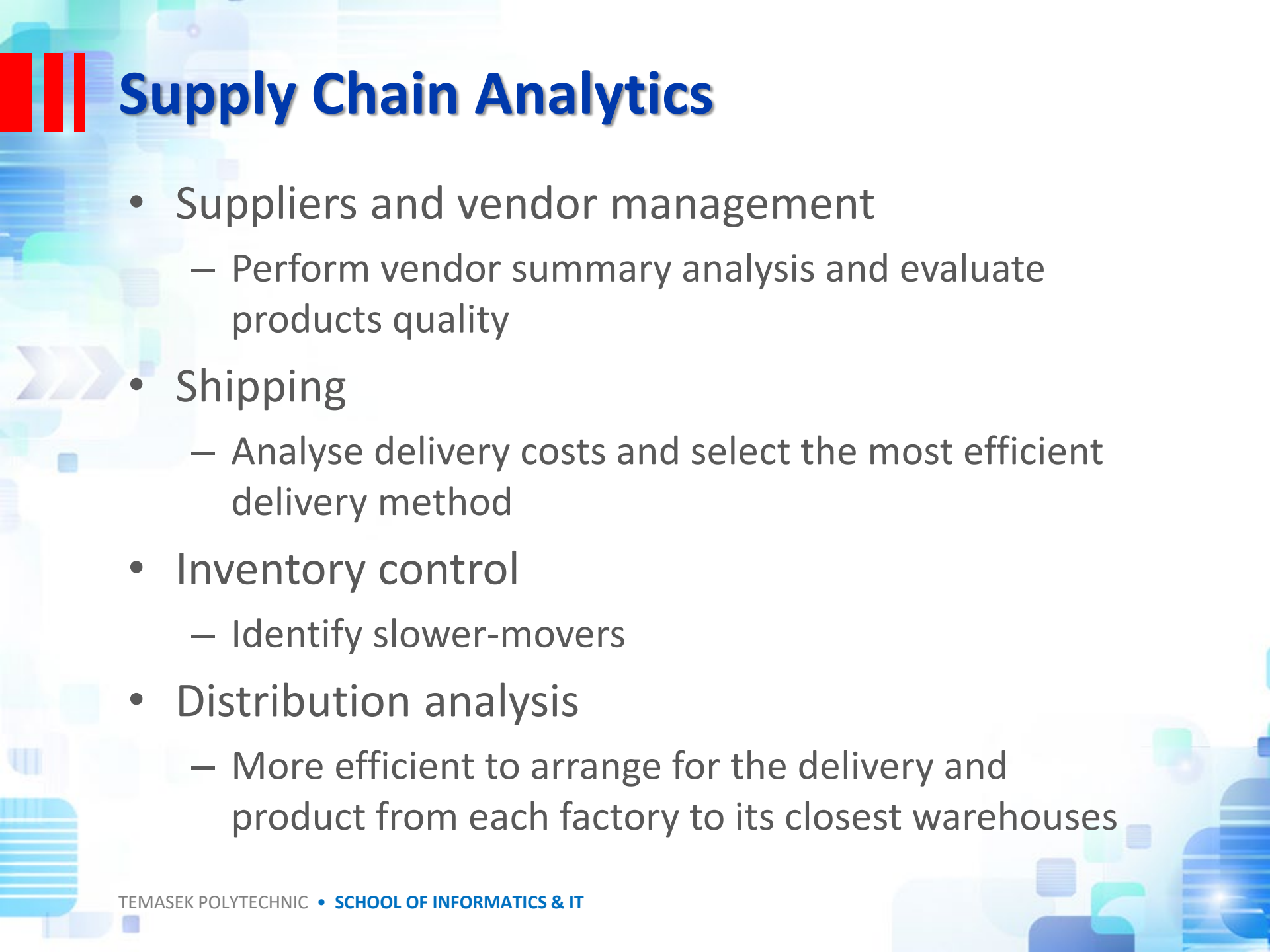
Business Productivity Analytics

- Defect analysis
 - Analyze factors that affect defective items, quality control of raw material
- Capacity planning and optimization
 - Understand resource utilization and assist management in resource planning and staffing
- Financial reporting
 - High-level analysis and the ability to drill down to particular area
- Risk management
 - Greater accuracy or precision in tracking business processes and productivity



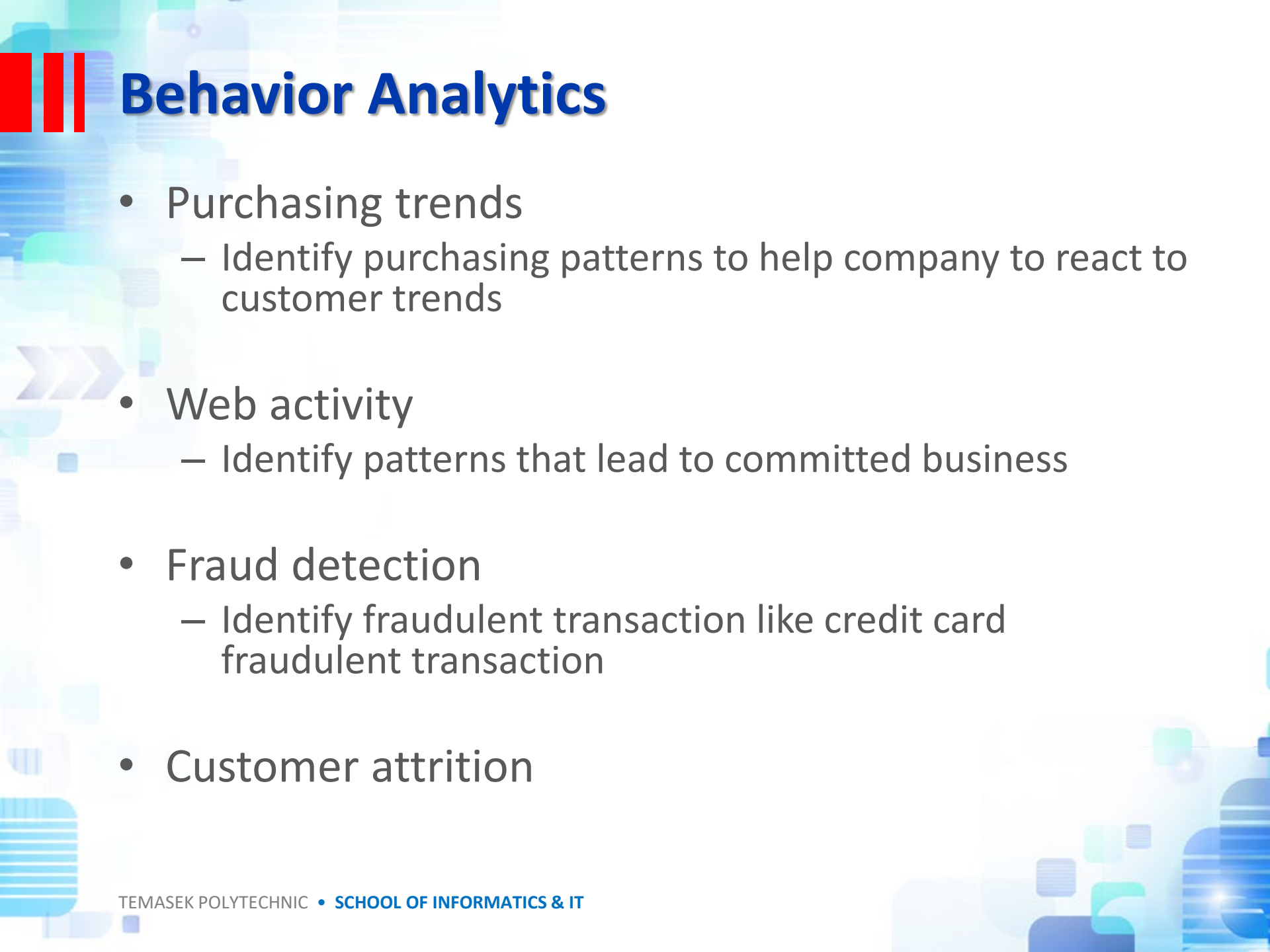
Sales Channel Analytics

- Marketing
 - Fine-tune marketing program to achieve higher marketing effectiveness
- Sales performance and pipeline
 - Analyse the data associated with the sales to identify which factors lead to higher sales, for example, region, sales profiles and contact frequency etc.



Supply Chain Analytics

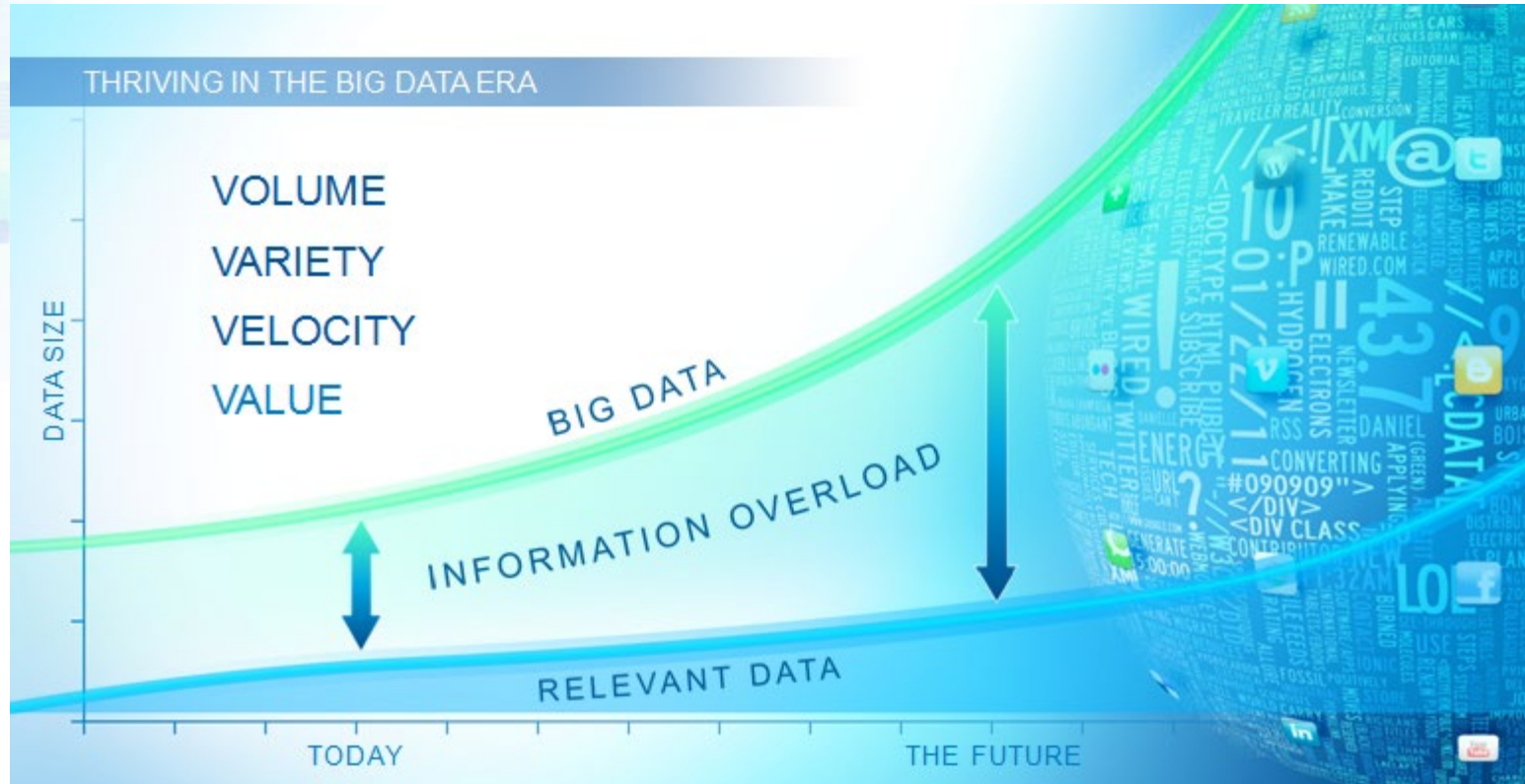
- Suppliers and vendor management
 - Perform vendor summary analysis and evaluate products quality
- Shipping
 - Analyse delivery costs and select the most efficient delivery method
- Inventory control
 - Identify slower-movers
- Distribution analysis
 - More efficient to arrange for the delivery and product from each factory to its closest warehouses



Behavior Analytics

- Purchasing trends
 - Identify purchasing patterns to help company to react to customer trends
- Web activity
 - Identify patterns that lead to committed business
- Fraud detection
 - Identify fraudulent transaction like credit card fraudulent transaction
- Customer attrition

Future Trends of BI



Top Business Intelligence Trends 2019 – BI Survey

The BI Survey is the world's largest annual survey of business intelligence (BI) software users.

- Data Discovery
- Self-Service BI
- Master Data/Data Quality Management
- Predictive/Advanced Analytics
- Data Preparation
- Analytical Databases
- Data Governance
- Lean Planning
- Integrated Platforms BI/PM
- Mobile BI

BI Software

Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms



Source: Gartner (February 2021)